

Pathology-Level ML Model Performance Summary (Test Set)
Classical ML models trained on neural network pathology predictions

Neural Network Mode	Pathology	ML Algorithm	AUC Mean	AUC Std	AUC 95% CI	F1 Mean	F1 Std	Optimal Threshold	Threshold Std	Samples	Positive Rate	Cross-Val Folds
HMV-MIL	A-lines	Random Forest	0.734	0.024	[0.704-0.763]	0.635	0.052	0.482	0.016	43689	0.472	5
HMV-MIL	A-lines	Logistic Regression	0.801	0.010	[0.789-0.813]	0.700	0.051	0.542	0.015	43689	0.472	5
HMV-MIL	A-lines	XGBoost	0.773	0.006	[0.765-0.781]	0.669	0.053	0.478	0.011	43689	0.472	5
HMV-MIL	Large Consolidation	Random Forest	0.803	0.057	[0.732-0.875]	0.354	0.108	0.414	0.023	43689	0.110	5
HMV-MIL	Large Consolidation	Logistic Regression	0.914	0.009	[0.903-0.926]	0.496	0.069	0.360	0.024	43689	0.110	5
HMV-MIL	Large Consolidation	XGBoost	0.855	0.038	[0.808-0.902]	0.457	0.055	0.160	0.019	43689	0.110	5
HMV-MIL	Pleural Effusion	Random Forest	0.648	0.053	[0.582-0.714]	0.084	0.032	0.264	0.057	43689	0.011	5
HMV-MIL	Pleural Effusion	Logistic Regression	0.773	0.027	[0.740-0.806]	0.053	0.017	0.544	0.026	43689	0.011	5
HMV-MIL	Pleural Effusion	XGBoost	0.800	0.052	[0.736-0.864]	0.070	0.015	0.038	0.008	43689	0.011	5
HMV-MIL B-lines and Small Consolidation	B-lines and Small Consolidation	Random Forest	0.621	0.008	[0.611-0.630]	0.503	0.046	0.470	0.019	43689	0.405	5
HMV-MIL B-lines and Small Consolidation		Logistic Regression	0.661	0.014	[0.644-0.678]	0.590	0.013	0.478	0.004	43689	0.405	5
HMV-MIL B-lines and Small Consolidation		XGBoost	0.660	0.022	[0.632-0.688]	0.560	0.047	0.434	0.009	43689	0.405	5
NoKeyframe	A-lines	Random Forest	0.754	0.064	[0.675-0.833]	0.681	0.048	0.464	0.023	43689	0.472	5
NoKeyframe	A-lines	Logistic Regression	0.790	0.036	[0.745-0.835]	0.719	0.038	0.504	0.021	43689	0.472	5
NoKeyframe	A-lines	XGBoost	0.790	0.054	[0.723-0.856]	0.696	0.055	0.480	0.010	43689	0.472	5
NoKeyframe	Large Consolidation	Random Forest	0.829	0.114	[0.687-0.971]	0.429	0.135	0.398	0.036	43689	0.110	5
NoKeyframe	Large Consolidation	Logistic Regression	0.908	0.030	[0.872-0.945]	0.488	0.143	0.512	0.039	43689	0.110	5
NoKeyframe	Large Consolidation	XGBoost	0.862	0.117	[0.716-1.007]	0.489	0.136	0.132	0.015	43689	0.110	5
NoKeyframe	Pleural Effusion	Random Forest	0.678	0.064	[0.599-0.757]	0.113	0.048	0.256	0.038	43689	0.011	5
NoKeyframe	Pleural Effusion	Logistic Regression	0.730	0.065	[0.648-0.811]	0.042	0.015	0.492	0.047	43689	0.011	5
NoKeyframe	Pleural Effusion	XGBoost	0.819	0.066	[0.737-0.900]	0.113	0.041	0.060	0.007	43689	0.011	5
NoKeyframe B-lines and Small Consolidation	B-lines and Small Consolidation	Random Forest	0.645	0.032	[0.606-0.685]	0.520	0.079	0.440	0.010	43689	0.405	5
NoKeyframe B-lines and Small Consolidation		Logistic Regression	0.672	0.031	[0.634-0.711]	0.492	0.275	0.488	0.004	43689	0.405	5
NoKeyframe B-lines and Small Consolidation		XGBoost	0.686	0.046	[0.629-0.743]	0.543	0.130	0.430	0.010	43689	0.405	5